Choose a motion method

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship and justify its assumptions.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A cart starts at 3 m and moves right at 5 m/s for 2 s. Predict final position. Allowed endpoints are 11 to 15 m; state this interval and one assumption.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_f = v_i + a\Delta t$; $\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$.
Constant acceleration; i = initial, f = final. At constant velocity, $\Delta x = v\Delta t$.

Worked reasoning: From rest, a cart accelerates uniformly at 2 m/s² for 3 s. Final velocity=6 m/s; average velocity=(0+6)/2=3 m/s; displacement=3×3=9 m.

Your turn: Show the same 9 m using the area of a triangle on a velocity-time graph.

Marzano check - circle: ① ② ③

03 Choose the relationship

A cart moves steadily at 3 m/s for 4 s. Which relationship fits: steady velocity or changing velocity? Find displacement.

Marzano check - circle: ① ② ③

04 Build final velocity

A cart starts at 2 m/s east. It gains 2 m/s each second for 3 s. Complete: gain=2×3=____ m/s; final=2+____=____ m/s east.

Time (s)	Velocity east (m/s)
0	2
1	4
2	6
3	?

Marzano check - circle: ① ② ③

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05 Find the distance while speeding up

Use the cart in 4. Its acceleration is constant. Average velocity=(2+8)÷2=____ m/s. Displacement=average velocity×3 s=____ m.

Marzano check - circle: 1 2 3

06 Choose a method

A cart travels west at 10 m/s and west at 3 m/s 2 s later. Is it speeding up? Find acceleration size and direction.

Marzano check - circle: 1 2 3

07 Exit check

A cart speeds up uniformly from 0 to 6 m/s east in 5 s. Find acceleration, average velocity and displacement. Show each step.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Create two different motions that travel 8 m in 4 s: one at steady speed and one speeding up uniformly from rest. Draw their velocity-time graphs.

Marzano check - circle: 1 2 3